GM UNIVERSITY

P.B. Road, Davanagere – 577 006, Karnataka, India

SCHOOL OF ENGINEERING | FACULTY OF ENGINEERING & TECHNOLOGY

Department of Electronics & Communication Engineering

PROJECT BASED LEARNING (PBL) REPORT

RAILWAY LEVEL CROSSING & TRAIN TRAFFIC CONTROLLER

Course Code: UE25EC2302 | **Course Title:** Hardware Modelling using Verilog

Semester: 3 | **Academic Year:** 2026–2027 | **Total Marks:** 40

SUBMITTED BY (TEAM MEMBERS):

Sl. No.	Student Name	USN / Roll No.	Project Responsibility
1	**Vikas K S (Lead)**	U26E02EC521	System Architecture, Top Hierarchy & FSM
2	**Nandan S L**	U26E02EC518	Gate-Level Safety Interlock & Failsafes
3	**Meghana P**	U25E02EC070	Traffic Lights, Buzzer PWM & LEDs
4	**Keerthana R F**	U25E02EC053	Testbench, Verification & Yosys Synthesis

Mr. Santhosh B G	**Dr. Praveen J**	**Dr. Praveen J**
Course Coordinator	Head of Department, ECE	Director, School of Engg.

1. ABSTRACT

Railway level crossings represent one of the most hazard-prone nodes in modern transport networks. Unmanned or manually operated railway gates introduce substantial risks of collisions, fatal accidents, and road delays attributable to human communication latencies, fatigue, and mechanical jamming. This report details the complete engineering design, Verilog HDL digital modelling, behavioral simulation, logic synthesis, and physical hardware prototype of an autonomous, fail-safe **Railway Level Crossing and Train Traffic Controller**.

The system automatically detects approaching and departing trains using debounced optical/inductive track sensors, coordinates motorized barrier gates through a 4-state Finite State Machine (FSM), regulates road traffic lights with pre-closure warning buffers, displays railway track signal aspects, and guarantees absolute priority for emergency events (track obstacles, manual emergency halts, and emergency vehicle preemption).

The digital architecture fulfills all six syllabus units of course **UE25EC2302 (Hardware Modelling using Verilog)**: Unit 1 top-down hierarchical design; Unit 2 gate-level modelling with built-in primitives (`and`, `or`, `not`, `buf`) enforcing an unbreachable hardware safety interlock; Unit 3 dataflow and behavioural control; Unit 4 reusable functions and multitasking (`fork...join`); Unit 5 parameters and system tasks; and Unit 6 logic synthesis via the **Yosys Open Synthesis Suite** resulting in 287 standard logic cells. The design was verified with **Icarus Verilog 12.0**, achieving a 100% pass rate across all test vectors with zero safety violations.

2. INTRODUCTION & BACKGROUND

According to global railway safety audits, accidents at grade crossings account for a disproportionate percentage of total railway fatalities. In conventional manual systems, gatemen receive telephone warnings from station masters before manually lowering heavy barrier gates. Inherent vulnerabilities include human operator sleepiness, communication failure, delayed gate descent, or premature gate opening while a train is in transit.

Implementing the control system directly in digital hardware (FPGA / ASIC) using **Verilog HDL** ensures:

- **Deterministic Real-Time Response:** Elimination of operating system latencies, context switches, or software crashes.
- **Fail-Safe Hardware Interlocks:** Direct combinational gate networks that physically inhibit gate opening commands when danger is present.
- **Silicon Synthesizability:** Clean mapping from RTL code to silicon standard cells or FPGA look-up tables (LUTs).

3. PROBLEM STATEMENT & COURSE OUTCOMES

> **Problem Statement (GM University UE25EC2302):**
> *"Design and develop a Railway Level Crossing and Train Traffic Controller using Verilog HDL that integrates concepts of digital modelling, behavioural control, tasks and functions, useful modelling techniques, and logic synthesis. The system must detect approaching and departing trains, control crossing gates automatically, generate traffic signals and buzzer indications, prioritize emergencies, and guarantee safe operation."*

Mapping to Course Learning Outcomes (COs):

CO Code	Course Outcome Description	Evaluation Component
CO2	Understand digital modelling principles and problem formulation	Problem Understanding (4 Marks)
CO3	Formulate design methodologies, state diagrams, and architectural blocks	Design & Methodology (5 Marks)
CO4	Implement structural, gate-level, dataflow, and behavioural models	Implementation Coverage (6 Marks)
CO5	Integrate hierarchical modules and verify system functionality	Functionality & Demo (8 Marks)
CO6	Demonstrate innovation, safety interlocking, teamwork, and documentation	Report Quality & Contribution (7 Marks)

4. METHODOLOGY & SYSTEM ARCHITECTURE

The system follows a modular, top-down hierarchical architecture divided into five interacting digital subsystems coordinated through handshaking and hardware interlocks:



4.1 Finite State Machine (FSM) State Transition Flow

Physical barrier motion is regulated by a synchronous Mealy/Moore FSM in `gate_controller.v`:

- **`ST_OPEN` (90°):** Default idle state. Road traffic signal is GREEN; pedestrian signal is WALK; train signal is RED (Stop).
- **`ST_CLOSING` (90° to 0°):** Train detected by Entry Sensor. Pre-closure warning initiates (`WARN_DELAY_CYCLES`). Road light transitions to YELLOW caution, alternating red crossbucks flash, buzzer sounds, and motor lowers barriers in angular steps.
- **`ST_CLOSED` (0°):** Barrier locked horizontally. Road light transitions to solid RED. Safety Interlock confirms barrier closure: train signal aspect turns **GREEN (PROCEED)**.
- **`ST_OPENING` (0° to 90°):** Train clears Exit Sensor. Post-clearance safety buffer (`POST_CLEAR_DELAY`) satisfies. Motor raises barrier to 90°, restoring road traffic to GREEN.

5. IMPLEMENTATION DETAILS (UNITS 1 TO 6)

5.1 Unit 1: Hierarchical Design (railway_level_crossing_top.v)

Realized through top-down modular instantiation using explicit named port connections (`.port(signal)`). Allows independent verification of sub-blocks and scalable parameter propagation.

5.2 Unit 2: Built-in Primitive Gate-Level Modelling (gate_safety_interlock.v)

To adhere to railway CENELEC SIL-4 safety regulations, critical gate locks cannot depend on software counters. A dedicated combinational network built purely from primitive gates (`and`, `or`, `not`, `buf`) arbitrates actuator commands:

```
  // Pure Gate-Level Modelling (Unit 2 Built-in Primitives)
  not u_inv_tp   (not_train_present, train_present);
  not u_inv_ta   (not_train_approaching, train_approaching);
  not u_inv_emg  (not_emergency, emergency_active);

  or  u_or_danger(danger_condition, train_present, train_approaching, emergency_active);
  and u_and_safe (normal_safe_permit, not_train_present, not_train_approaching, not_emergency);
  and u_and_open (interlock_open_en, req_open, normal_safe_permit);
  or  u_or_close (interlock_close_en, req_close, danger_condition);
  buf u_buf_lock (barrier_lock_cmd, danger_condition);
  and u_and_alarm(safety_violation_alarm, manual_force_open, danger_condition);
```

5.3 Unit 3: Dataflow and Behavioural Modelling

Dataflow: Used in `traffic_light_controller.v` via continuous assignments (`assign`):

```
  assign road_green         = (gate_is_open) & (~gate_warning) & (~train_approaching) & (~train_present) &
(~emergency_active);
  assign road_yellow        = (gate_warning || (gate_state == 2'b01)) & (~emergency_active);
  assign road_red           = (gate_state == 2'b10) || (gate_state == 2'b11) || emergency_active || (gate_warning &&
blink_pulse);
  assign train_signal_green = (gate_is_closed) & (~obstacle_detected) & (~emergency_active);
  assign train_signal_red   = ~train_signal_green;
```

Behavioural: Procedural edge-triggered blocks (`always @(posedge clk or negedge rst_n)`) model the sequential debouncing shift-registers and FSM state registers.

5.4 Unit 4: Tasks, Functions & Multitasking Concepts

- **Function:** `check_sensor_validity` in `train_detection.v` validates sequence consistency, detecting sensor sequence anomalies in zero simulation time.
- **Task:** `update_barrier_position` in `gate_controller.v` models motor angular displacement; testbench tasks (`simulate_normal_train_passage`) encapsulate stimulus routines.
- **Multitasking:** `fork...join` in `railway_level_crossing_tb.v` models concurrent asynchronous train traversal and emergency vehicle requests simultaneously.

5.5 Unit 5: Useful Techniques, Parameters & System Tasks

Parameterized constants (`DEBOUNCE_CYCLES`, `WARN_DELAY_CYCLES`), localparams for safe state encodings, conditional priority encoders, and simulation system tasks (`$dumpfile`, `$monitor`, `$time`, `$display`, `$finish`).

5.6 Unit 6: Logic Synthesis & Gate-Level Verification

RTL code was synthesized to standard cell gates using **Yosys 0.52** via script `synth.ys`, proving 100% synthesizability without inferred latches or asynchronous feedback loops.

6. RESULTS & VERIFICATION DISCUSSION

Simulation was conducted in **Icarus Verilog 12.0** and visualized using GTKWave. Five automated test cases verified all operational modes:

Test ID	Test Scenario	Expected Safety Behaviour	Observed Hardware Status	Verdict
TC-1	System Reset Baseline	Gate 90° Open, Road Green, Train Aspect Red	Gate 90°, Road Green, Buzzer Silent	PASS
TC-2	Normal Train Passage	Warning → Close → Train Pass → Safe Open	Full 90° → 0° → 90° angular transit	PASS
TC-3	Safety Interlock Tamper	Hardware gate primitives block manual open request	`interlock_open_en=0`, Alarm trips	PASS
TC-4	Track Obstacle (Stalled Car)	Train signal forced RED; emergency siren sounds	Train halted, emergency active	PASS
TC-5	Multitasking Concurrency	Asynchronous emergency vehicle preempts train	Immediate emergency priority verified	PASS

Logic Synthesis Cell Counts (Yosys Open Synthesis Suite 0.52):

Standard Logic Cell Type	Yosys Gate Primitive	Instance Count	Functional Role
Logic AND Gates	`$_AND_`	60	Interlocks, permit gating, decoders
Logic NAND Gates	`$_NAND_`	59	Optimized combinational logic
Logic NOR / NOT Gates	`$_NOR_` / `$_NOT_`	32 / 25	Active-low decoders & inverters
Logic OR / XOR / XNOR	`$_OR_` / `$_XOR_` / `$_XNOR_`	22 / 10 / 3	Priority combiners & parity filters
Multiplexers (MUX)	`$_MUX_`	4	Data selection
Sequential Flip-Flops	`$_DFF_PN0_` / `$_DFFE_PN0P_`	72	Registers, timers & FSM states
TOTAL SYNTHESIZED STANDARD CELLS		**287 Cells**	**100% Synthesizable RTL (Zero Inferred Latches)**

7. PHYSICAL HARDWARE PROTOTYPE CONSTRUCTION & CIRCUIT SCHEMATIC

To satisfy the working model demonstration requirement, a tabletop miniature physical diorama was constructed with the complete electrical circuit schematic depicted below:



- **TowerPro SG90 Micro Servos (2x):** Actuate barrier arms across the two-lane roadway (90° ↔ 0°). Powered by dedicated external 5V 2A DC supply with shared common ground.
- **LM393 Infrared Proximity Sensors (4x):** Track Entry (D2), Track Occupancy (D3), Track Departure (D7), and Roadway Obstacle (A1).
- **Traffic & Warning Cluster:** Road Signals (D4 Red, D5 Yellow, D6 Green via 220Ω), Crossbuck Wig-Wags (D12, D13), Track Signals (A2 Green, A3 Red), and 5V active piezo buzzer (D8).
- **Failsafe Grounding:** Microcontroller GND and Servo Power Supply GND tied into a unified common ground rail.

8. CONCLUSION & FUTURE SCOPE

The **Railway Level Crossing and Train Traffic Controller** successfully proves that dedicated digital hardware modelling eliminates human operational latencies and accident risks at railway level crossings. Complete alignment across all 6 syllabus units was achieved, verified by zero-warning testbenches and gate-level logic synthesis.

Future Scope: Integration of edge-AI computer vision for obstacle classification, wireless IoT LoRaWAN telemetry for centralized train dispatch monitoring, and solar-charged supercapacitor battery backup for failsafe barrier descent during grid failure.

9. REFERENCES

1. Samir Palnitkar, *"Verilog HDL: A Guide to Digital Design and Synthesis"*, 2nd Edition, Prentice Hall PTR, 2003.
2. Stephen Brown and Zvonko Vranesic, *"Fundamentals of Digital Logic with Verilog Design"*, 3rd Edition, McGraw-Hill, 2013.
3. Michael D. Ciletti, *"Advanced Digital Design with the Verilog HDL"*, 2nd Edition, Pearson Education, 2011.

4. Indian Railways Telecommunication Manual (IRTM), *"Standard Signaling & Interlocking Specifications for Level Crossing Gates"*, Ministry of Railways, Government of India.
5. IEEE Std 1364-2005, *"IEEE Standard for Verilog Hardware Description Language"*, IEEE Computer Society, 2006.
6. Yosys Open SYnthesis Suite Documentation, YosysHQ GmbH (https://yosyshq.net/yosys/).

10. INDIVIDUAL CONTRIBUTION & EVALUATION RUBRICS (40 MARKS)

Evaluation Component	Marks	CO Mapping	Assigned Team Member	Demonstrated Contribution
Problem Understanding	4	CO2	Vikas K S	System requirements, safety constraints & specifications
Design & Methodology	5	CO3	Vikas K S	Architectural partitioning, FSM flowcharts & block diagram
Implementation (All Units)	6	CO4	All Members	Units 1 to 6 Verilog source coding & structural primitives
Functionality & Integration	5	CO5	Nandan S L	Submodule integration & hardware interlock arbitration
Innovation / Enhancements	3	CO6	Meghana P	Web Audio SCADA cockpit, LiDAR obstacle detection & LEDs
Report Quality	2	CO6	Keerthana R F	Publication-quality IEEE documentation & waveform graphs
Demonstration	3	CO5	All Members	Live physical diorama, servo actuators & test vectors
Viva Voce	2	CO4	Individual	Verilog concepts, blocking vs non-blocking & synthesis
Individual Contribution	10	CO3,4,6	All Members	Technical tasks, documentation, and equal 25% share
TOTAL EVALUATION MARKS	**40 / 40**	**CO2–CO6**	**Comprehensive Project Submission**	

INDIVIDUAL CONTRIBUTION DECLARATION TABLE

Name	Role	Contribution Details	% Share	Signature
Vikas K S	Team Lead	Architecture, Top Module & FSM Controller	**25%**	_____
Nandan S L	Safety Specialist	Gate-Level Interlock & Primitive Circuits	**25%**	_____
Meghana P	Actuator Lead	Traffic Signals, Buzzer PWM & Flasher LEDs	**25%**	_____
Keerthana R F	Verification Lead	Testbench Tasks, Verification & Yosys Synthesis	**25%**	_____

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